YMNOTES MEDICAL

Peritoneum, Omentum & Mesentery

Kasr El-Aini Introduction to Surgery • Ch. 35 (Expanded Edition)

REDESIGNED EDITION

About this edition

SOURCE

Redesigned from 'Peritoneum, Omentum & Mesentery (Expanded)' slide deck — Kasr El-Aini Introduction to Surgery, Ch. 35 (pp. 446–457), Ayman Salah & Omar El-Kashef. Reordered into Mid-Module and After Mid-Module parts.

WHAT CHANGED

Presentation only. The medical text is reproduced from the original book: typography, hierarchy, page layout, tables, figures and navigation were rebuilt; the wording, numbers, doses, units, thresholds, classifications and recommendations were not.

WHAT WAS CORRECTED

Any evidenced change to supplied wording is identified in the correction register with its original form. Uncertain source readings remain explicitly unresolved.

EDITORIAL ADDITIONS

Chapter grouping, navigation labels, figure numbers and source-page references were added by YMnotes for navigation. Editorial notes are shown in a distinct panel so they can never be mistaken for the source book's own words.

FIDELITY

The medicine in this edition is the medicine in the original book. Nothing has been added, removed, reworded or summarised; only the presentation is new. Where the original was demonstrably defective, the repair is marked on the page and listed in the editorial corrections.

REPAIRED SOURCE DEFECT

Where the supplied source cannot establish a character or reading, uncertainty is retained. A source restoration must be evidenced by the original material, not inferred from medical knowledge.

OTHER CORRECTIONS

Typographical defects in the source — misspelled drug names, malformed medical terms, lost word spaces — were corrected where the evidence was unambiguous, and marked where the correction changes a word. Nothing was rewritten for style, and no medical statement was modernised: where the source is dated rather than wrong, it stands as written.

Contents

01 PART 1 · MID-MODULE: Peritonitis & Subphrenic Abscess 1
Surgery · Mid-Module

02 PART 2 · AFTER MID-MODULE: Peritoneum, Omentum & Mesentery 7
Surgery · After Mid-Module

How to read this book

CHAPTER OPENER

Each chapter opens on its own page with the chapter number and the original book's own title for that material.

TOPIC BAND

A tinted band with a heavy left rule marks a major topic. The running header repeats the current topic on every page.

CONDITION

A named disease, drug or entity sits under a lighter rule directly beneath its topic.

RECURRING SECTIONS

Definition, Causes, Clinical Picture, Investigations, Treatment and Complications carry a small accent marker. Treatment runs wide and dark; complications and contraindications run red; investigations run blue.

BULLET DEPTH

A filled diamond is a first-level point, a filled square a second, a short dash a third and a hollow circle a fourth. Shape carries the depth — not size or colour — so the nesting survives greyscale printing and reads at arm’s length.

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A dashed violet panel is written by YMnotes, not by the original book. It explains organisation only and adds no medical content.

EDITORIAL CORRECTION

A pale amber word, symbol or number is one YMnotes verified and corrected or restored from the supplied source. The original reading is recorded for every one of them.

SOURCE PAGES

Figure captions cite the page of the original book they came from, so any passage can be traced back to the source.

YMnotes

CHAPTER 01

PART 1 · MID-MODULE: Peritonitis & Subphrenic Abscess

01

IN THIS CHAPTER

01 Microorganisms & Routes of Infection
02 Pathology & Fate of Peritonitis
03 Clinical Picture of Generalized Peritonitis
04 Investigations & Surgical Management
05 Iliac Abscess: Aetiology & Management
06 Pelvic Abscess: Clinical & Drainage Routes
07 Seven Anatomical Subphrenic Spaces
08 Subphrenic Abscess: Signs & Percussion
09 Subphrenic Imaging & Surgical Drainage
10 Golden Diagnostic Reminders

Microorganisms & Routes of Infection

SECTION 02 • GENERALIZED SEPTIC PERITONITIS

Pyogenic Pathogens:

Escherichia coli, aerobic/anaerobic Streptococci, Bacteroides, Pneumococci, Staphylococci.

- **1. Local Spread**

Appendicitis cholecystitis ulcer perforation anastomotic leak.

- **2. External Entry**

Penetrating trauma wounds or surgical incisions.

- **3. Haematogenous**

Rare blood-borne seeding during systemic septicaemia.

- **4. Primary Route**

Fallopian tube ascent in young girls.

Pathology & Fate of Peritonitis

SECTION 02 • PATHOPHYSIOLOGY & OUTCOME

- **Gross Pathological Changes**
  - **Fibrinous Exudation:** Opaque membranes adhere via fibrin plaques.
  - **Purulent Collection:** Excessive free pus inside peritoneal cavity.
  - **Paralytic Ileus:** Protective neural reflex aggravated by toxins.
  - **Mechanical Obstruction:** Late progression from dense fibrinous adhesions.
- **Clinical Course & Flaring Factors**
  - **Resolution or Abscess:** Walled off by omentum or recesses.
  - **Peristalsis Stimulation:** Oral intake enemas or laxatives flare sepsis.
  - **Host Vulnerability:** Diabetes steroids AIDS small childhood omentum.
  - **Decompensation:** Unchecked septicaemia multiorgan failure death.

Clinical Picture of Generalized Peritonitis

SECTION 02 • BEDSIDE DIAGNOSIS

- **Cardinal Symptoms**
  - **Abdominal Pain:** Constant ache worsened by movement or cough.
  - **Progressive Vomiting:** Gastric then bile-stained then faeculent.
  - **Abdominal Distension:** Progressive swelling from ileus and fluid.

- **Bowel Habit:** Absolute constipation or pelvic tenesmus.

Physical Examination Signs

- **General State:** Motionless toxic patient pyrexia tachycardia.
- **Palpation:** Diffuse tenderness rebound board-like muscular rigidity.
- **Auscultation:** Silent abdomen from paralytic ileus.
- **Late Neglected Stage:** Sunken eyes Facies hippocratica septic shock.

Investigations & Surgical Management

SECTION 02 • WORKUP & OPERATIVE PROTOCOL

Operative Details:

Midline incision, pus culture, source elimination, warm saline lavage, peritoneal + subcutaneous drains, ICU care.

01 • Workup

Leucocytosis subdiaphragmatic gas ultrasound aspiration.

02 • Pre-Op Care

NG decompression IV fluids Foley catheter.

03 • Antibiotics

Triple IV ampicillin aminoglycoside metronidazole.

04 • Laparotomy

Source control saline toilet peritoneal drains.

Iliac Abscess: Aetiology & Management

SECTION 03 • LOCALIZED SEPTIC PERITONITIS (PART A)

Anatomical Aetiology

- **Right Iliac Fossa:** Appendicitis or perforated duodenal ulcer.
- **Left Iliac Fossa:** Diverticulitis or perforated sigmoid carcinoma.
- **Bilateral Causes:** Pelvic infection spread or resolving peritonitis.

Clinical & Diagnostic

- **Symptoms:** Iliac pain swelling hectic fever vomiting.
- **Local Signs:** Tenderness rigidity overlying skin erythema.
- **Workup:** Leucocytosis plus ultrasound volume estimation.

Surgical Treatment

- **Open Drainage:** Extraperitoneal muscle-cutting incision.
- **Guided Drainage:** Percutaneous ultrasound or CT catheter.
- **Appendicular Rule:** Interval appendicectomy 6 months later.

Pelvic Abscess: Clinical & Drainage Routes

SECTION 03 • LOCALIZED SEPTIC PERITONITIS (PART B)

Pathogenesis & Clinical Features

- **Anatomical Pouch:** Recto-vesical (males) or Douglas pouch (females).
- **Three Causes:** Pelvic appendicitis resolving peritonitis or PID.
- **Rectal Irritation:** Tenesmus and pathognomonic mucous diarrhoea.
- **Bladder Irritation:** Dysuria and frequent micturition.
- **Digital Rectal Exam:** Tender boggy anterior rectal fullness.

Drainage Protocols & Caveat

- **Transrectal Drainage:** Indicated when pointing into rectum.
- **Posterior Colpotomy:** Drainage through posterior vaginal fornix.
- **Suprapubic Route:** Extraperitoneal drainage for suprapubic pointing.
- **Surgical Caveat:** Prefer transrectal route to avoid peritoneal spread.

Seven Anatomical Subphrenic Spaces

SECTION 03 • LOCALIZED SEPTIC PERITONITIS (PART C)

Right-Sided Compartments (3)

- **1. Right Suprahepatic:** Diaphragm to right lobe medial to falciform.
- **2. Right Infrahepatic (Morison):** Liver to right kidney and duodenum.
- **3. Right Extraperitoneal:** Liver bare area to diaphragm.

Left-Sided Compartments (4)

- **4. Left Suprahepatic:** Left diaphragm above left liver and spleen.
- **5. Left Anterior Infrahepatic:** Left liver to stomach and lesser omentum.
- **6. Left Posterior Infrahepatic:** Anatomically equals the lesser sac.
- **7. Left Extraperitoneal:** Surrounds upper left kidney and adrenal.

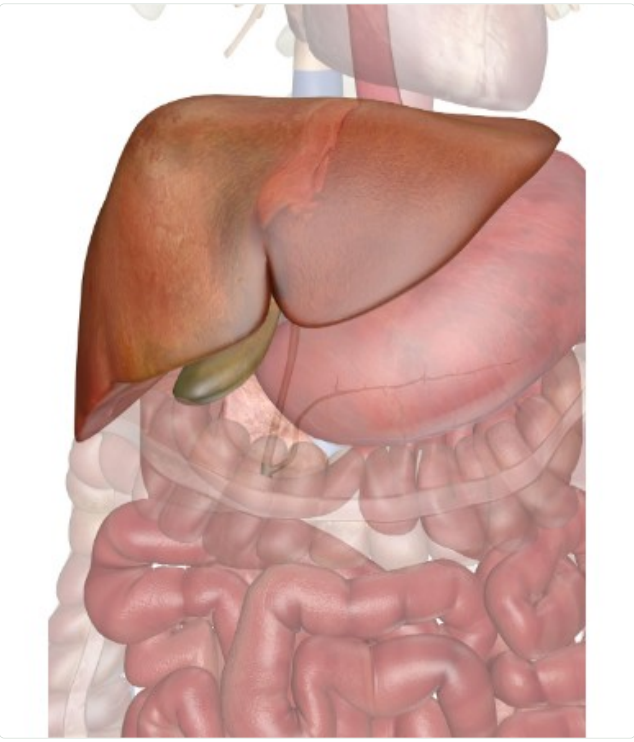


Fig 1.1 — original page 3

Subphrenic Abscess: Signs & Percussion

SECTION 03 • SUBPHRENIC CLINICAL PRESENTATION

Systemic & Local Physical Signs

- **Systemic Toxaemia:** Hectic fever sweats wasting tachycardia.
- **Phrenic Irritation:** Persistent hiccups ipsilateral shoulder pain.
- **Inspection & Palpation:** Reduced chest movement intercostal tenderness.
- **Organ Shift:** Downward liver displacement upward apex beat.

Four Vertical Percussion Zones

- **Zone 1 (Superior):** Normal upper lung resonance.
- **Zone 2 (Upper-Mid):** Dullness from reactive pleural effusion.
- **Zone 3 (Lower-Mid):** Resonance/tympany from gas in abscess.
- **Zone 4 (Inferior Base):** Dullness from liver and liquid pus.

Subphrenic Imaging & Surgical Drainage

SECTION 03 • SUBPHRENIC WORKUP & EXTRASEROUS SURGERY

Radiology & Imaging

- **Diaphragm Sign:** Elevated thickened immobile tented dome.
- **Pleural Angle:** Obliterated costophrenic angle effusion.
- **Abscess Cavity:** Subdiaphragmatic air-fluid level on CT/US.

Ochsner's Approach

- **Indication:** Posterior subphrenic abscess collections.
- **Incision:** Subperiosteal 12th rib resection at L1 bed.
- **Plane:** Extrapleural extraperitoneal finger dissection above kidney.

Clairmont & Guided

- **Clairmont Route:** Anterior subcostal extraperitoneal dissection.
- **Percutaneous:** Ultrasound or CT-guided catheter drainage.
- **Conversion Rule:** Open drainage for thick multilocular pus.

Golden Diagnostic Reminders

Pelvic Abscess

Post-peritonitis fever mucous diarrhoea dysuria.

Subphrenic Abscess

Postoperative hectic fever without local signs.

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CHAPTER 02

PART 2 · AFTER MID-MODULE: Peritoneum, Omentum & Mesentery

02

IN THIS CHAPTER

01 Peritoneal Divisions, Pain & Healing
02 TB Spread & Five Pathological Varieties
03 TB Peritonitis: Diagnosis & Treatment
04 Ascites: Clinical Threshold & Aetiology
05 Differential Diagnosis of Ascites
06 Primary & Secondary Peritoneal Tumours
07 Omental Torsion & Mesenteric Cysts
08 Tillaux's Triad & Cyst Differential
09 Non-Specific Mesenteric Lymphadenitis
10 Tabes Mesenterica: Pathology & Care
11 Retroperitoneal Anatomy & Tumours
12 Golden Diagnostic Reminders

Peritoneal Divisions, Pain & Healing

SECTION 01 • ANATOMICAL & PHYSIOLOGICAL FACTS

Peritoneal Healing Rule:

Simultaneous in-situ metamorphosis of mesenchymal cells into mesothelium allows huge defects to heal rapidly.

- **Parietal Peritoneum**
  - **Anatomical Lining:** Inner abdominopelvic walls inside fascia transversalis.
  - **Nerve Supply:** Somatic plus visceral afferent fibers.
  - **Pain Character:** Sharp well-localized somatic pain.
  - **Motor Reflex:** Voluntary guarding then reflex muscle spasm.
- **Visceral Peritoneum**
  - **Anatomical Lining:** Invests intra-abdominal viscera and mesenteries.
  - **Nerve Supply:** Autonomic nerve fibers exclusively.
  - **Pain Character:** Dull poorly localized visceral pain.
  - **Histology:** Single mesothelial layer over loose connective tissue.

TB Spread & Five Pathological Varieties

SECTION 04 • TUBERCULOUS PERITONITIS (PART A)

Secondary Pathways:

Direct spread (mesenteric nodes, ileocaecal enteritis, salpingitis), haematogenous (pulmonary TB), or lymphatic.

- **1. Acute Miliary**

Serosa tubercles mimics acute peritonitis.

- **2. Caseous Type**

Cold abscesses sinuses internal fistulae.

- **3. Ascitic Type**

Commonest straw fluid sausage omentum.

- **4. Encysted**

Walled-off cyst mimics ovarian cyst.

- **5. Adhesive**

Dense dry adhesions bowel obstruction.

TB Peritonitis: Diagnosis & Treatment

SECTION 04 • TUBERCULOUS PERITONITIS (PART B)

Clinical Picture (Children & Young Adults)

- **TB Toxaemia:** Night fever drenching sweats anorexia wasting.
- **Doughy Abdomen:** Matted nodes tuberculomas mild guarding.
- **Rolled Omentum:** Transverse sausage mass above umbilicus.
- **Pelvic Exam:** May detect tubo-ovarian adnexal mass.

Investigations & Medical Care

- **Blood & Skin:** Anaemia lymphocytosis high ESR positive tuberculin.
- **Paracentesis:** Straw exudate (>1020) lymphocytes culture/inoculation.
- **Laparoscopy:** Most reliable targeted peritoneal biopsy proof.
- **Treatment:** Anti-TB drugs ≥1 year; surgery for obstruction.

Ascites: Clinical Threshold & Aetiology

SECTION 05 • PATHOLOGICAL PERITONEAL FLUID

1500 ml

Minimum free peritoneal fluid volume required for clinical detection on physical examination.

1. General Causes (Transudate + Generalized Oedema)

Liver cirrhosis congestive heart failure nephrotic syndrome severe hypoproteinaemia.

2. Local Causes (Exudate)

Tuberculous peritonitis visceral malignancy chylous ascites pancreatic ascites.

3. Rare Syndromic Causes

Meigs' syndrome (benign ovarian fibroma + pleural effusion) and Pseudomyxoma peritonii.

Differential Diagnosis of Ascites

SECTION 05 • TABLE 35.1 COMPLETE DIFFERENTIAL

Feature	Cirrhotic Ascites	Malignant Ascites	Tuberculous Ascites
Age & History	Any age; jaundice haematemesis	Elderly; short primary history	Young; chronic fever weight loss
General Exam	Hepatic insufficiency stigmata	Distant metastases evidence	Low-grade pyrexia cachexia
Abdominal Exam	Hepatosplenomegaly	Multiple hard irregular masses	Doughy abdomen matted masses
LFTs & Ultrasound	Impaired LFTs; shrunken liver	Normal LFTs; tumour masses	Normal LFTs; loculated fluid
Ascitic Fluid Tap	Transudate (<1020)	Blood-stained exudate (>1020)	Straw exudate (>1020) TB culture
Treatment	Medical; peritoneo-venous shunt	Tapping; radioactive gold	Anti-TB chemotherapy
Prognosis	Poor ("Bad")	Dismal ("Very bad")	Favourable ("Good")

Primary & Secondary Peritoneal Tumours

SECTION 06 • PERITONEAL NEOPLASMS

Carcinoma Peritonii

- **Origin:** Transcoelomic seeding from GI breast ovary.
- **Pathology:** Hemorrhagic ascites nodules frozen pelvis.
- **Palliation:** Paracentesis plus radioactive gold instillation.

Pseudomyxoma Peritonii

- **Aetiology:** Ruptured appendiceal mucocele or ovarian cyst.
- **Clinical Sign:** Yellow jelly masses without shifting dullness.
- **Surgery:** Scoop out jelly excise appendix/ovary.

Peritoneal Mesothelioma

- **Aetiology:** Rare primary malignancy linked to asbestos.
- **Diffuse Form:** Presents with exudative ascitic fluid.
- **Localized Form:** Bulky palpable intra-abdominal tumour mass.

Omental Torsion & Mesenteric Cysts

SECTIONS 07 & 08 • OMENTAL & MESENTERIC LESIONS

Torsion of the Omentum

- **Mechanism:** Longitudinal twisting from distal hernia adhesion.
- **Presentation:** Sudden acute pain tenderness muscular rigidity.
- **Diagnosis:** Established intraoperatively during laparotomy.
- **Treatment:** Surgical resection of strangulated omentum.

Mesenteric Cyst Classification & Surgery

- **False Cysts (No Lining):** Traumatic haematoma (evacuate) or TB cold abscess (medical).
- **Chylolymphatic (Commonest True):** Thin wall milky chyle separate vessels → enucleation.
- **Enterogenous (True):** Thick mucosal wall shared blood supply → bowel resection.
- **Dermoid & Hydatid:** True cysts treated by surgical enucleation.

Tillaux's Triad & Cyst Differential

SECTION 08 • CLINICAL SIGNS OF MESENTERIC CYSTS

Tillaux's Classic Clinical Triad

- **1. Swelling Site:** Painless cystic mass near umbilicus.
- **2. Axis Mobility:** Mobile perpendicular to mesenteric root only.
- **3. Percussion:** Dull cyst crossed by resonant bowel band.

Large Abdominal Cyst Differential

- **Pancreatic Pseudocyst:** Occupies upper abdominal compartment.
- **Ovarian Cyst:** Occupies pelvi-abdominal compartment.
- **Hydronephrosis:** Extends posteriorly into renal loin angle.

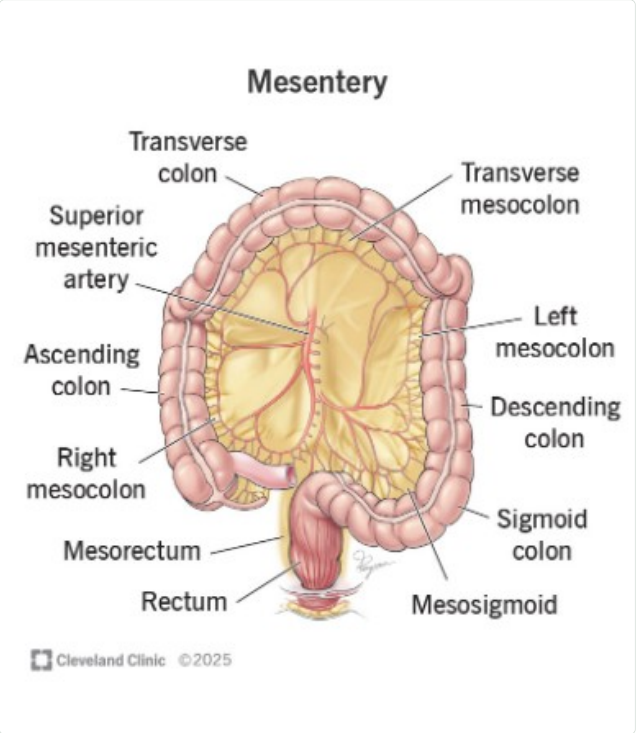


Fig 2.1 — original page 7

Non-Specific Mesenteric Lymphadenitis

SECTION 09 • PEDIATRIC ACUTE ABDOMEN (TABLE 35.2)

Pathogenesis:

Post-viral URI in children; reactive ileocaecal node hyperplasia + clear serous fluid; voluntary guarding without true rigidity.

Clinical Feature	Acute Appendicitis	Non-Specific Mesenteric Adenitis
Course of Pain	Continuous progressive pain	Complete pain freedom between colicky attacks
Maximal Tenderness	Fixed at McBurney's point	Above and medial to McBurney's point
Rovsing's Sign	Positive	Negative
Shifting Tenderness (Klein's)	Absent	Present (shifts medially in left decubitus)

Tabes Mesenterica: Pathology & Care

SECTION 09 • TUBERCULOUS MESENTERIC ADENITIS

Pathogenesis & Pathological Fate

- **Bovine Milk Focus:** Ileal Peyer's patch lesion + massive mesenteric nodes.
- **Resolution Pathway:** Dystrophic calcification fibrosis bowel kinking.
- **Caseous Activation:** Forms tuberculous mesenteric cold abscess.
- **Dissemination:** Node rupture causes TB peritonitis or miliary TB.

Clinical Features, Workup & Therapy

- **Symptoms:** Fever sweats wasting pallor pain diarrhoea.
- **Abdominal Signs:** Tender irregular matted nodes below/right of umbilicus.
- **Investigations:** Normal WBC positive tuberculin mottled X-ray calcifications.
- **Treatment:** Anti-TB drugs for active disease; none if calcified.

Retroperitoneal Anatomy & Tumours

SECTION 10 • RETROPERITONEAL SPACE & NEOPLASMS

Surgical Boundaries

- **Anterior/Posterior:** Parietal peritoneum to spine and psoas.
- **Superior/Inferior:** 12th rib diaphragm to pelvic brim.
- **Clinical Rule:** Zero internal barriers allow unimpeded spread.

Retroperitoneal Sarcoma

- **Disproportion:** Massive fixed hard mass mild symptoms.
- **Infiltration:** Colonic obstruction or ureteric hydronephrosis.
- **Management:** CT staging exploration biopsy + radiotherapy.

Lipoma & Lymphoma

- **Lipoma:** Enormous slow-growing mass in females.
- **Malignant Risk:** Myxomatous degeneration into liposarcoma.
- **Lymph Nodes:** Enlargement chiefly caused by malignant lymphoma.

Golden Diagnostic Reminders

Tuberculous Peritonitis

Emaciated youth chronic fever doughy abdomen.

Image Sources

- https://innerbody.imgix.net/Liver.png
- **Source:** www.innerbody.com
- https://my.clevelandclinic.org/-/scassets/Images/org/health/articles/mesentery.jpg
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